

Software Requirement Specification (SRS)

Project: College Campus Automation – Food & Dining Module

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Table of Contents

1. Introduction 1.1. Purpose 1.2. Project Scope 1.3. Definitions, Acronyms, and Abbreviations
 2. Overall Description 2.1. Product Perspective 2.2. User Classes and Characteristics 2.3. Operating Environment 2.4. Design and Implementation Constraints
 3. System Features (Functional Requirements) 3.1. User Authentication & Management 3.2. Student & Faculty Features 3.3. Canteen Administrator Features 3.4. Mess Administrator Features 3.5. Food Wastage Tracker
 4. External Interface Requirements 4.1. User Interfaces (UI) 4.2. Software Interfaces (API) 4.3. Hardware Interfaces 4.4. Communications Interfaces
 5. Non-Functional Requirements 5.1. Performance 5.2. Scalability 5.3. Security 5.4. Reliability 5.5. Usability
 6. System Architecture & Data Flow 6.1. Technology Stack 6.2. API Endpoint Specification 6.3. Database Schema 6.4. Payment Gateway Integration Flow
 7. Future Enhancements (Real-World Features)
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1. Introduction

1.1 Purpose

This Software Requirement Specification (SRS) document provides a complete blueprint for the “Food & Dining” module of the College Campus Automation system. It details the system’s scope, functionalities, and technical specifications, including the newly introduced features for **OTP-based order verification**, **free delivery**, and **promotional offers**.

1.2 Project Scope

The project entails the design, development, and deployment of a multi-tenant, scalable web application to modernize food services. The scope covers:

- **Canteen Pre-Ordering:** Online ordering, payment, and **free delivery** with **OTP verification**.
- **Digital Mess Management:** QR code-based meal redemption.
- **Democratic Menu Voting:** Student/faculty voting on upcoming mess menus.
- **Promotional Offers:** System for creating and applying discounts/offers.
- **Food Wastage Tracker:** Real-time dashboard for waste monitoring.

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
SRS	Software Requirement Specification
MERN	MongoDB, Express.js, React.js, Node.js
API	Application Programming Interface
JWT	JSON Web Token
OTP	One-Time Password (for order verification)
PRN	Permanent Registration Number (for students)
Admin	Canteen or Mess Operator

2. Overall Description

2.1 Product Perspective

The module is a core component of the campus automation platform, designed as a scalable, API-first web service. It will be built to handle the high transaction volume expected from a campus-wide food service system.

2.2 User Classes and Characteristics

User Class	Characteristics
Student/Faculty	End-users who need a fast, mobile-friendly interface to order food, apply offers, vote, and receive delivery.
Canteen Admin	Staff who manage menu, offers, inventory, process orders, and verify delivery using OTP.
Mess Admin	Staff who manage mess menu voting, finalize menus, and track meal redemptions.
Super Admin	System owner responsible for multi-college onboarding and platform management.

2.3 Operating Environment

Cloud-hosted web application, accessible via modern web browsers on all devices. Requires a stable internet connection.

2.4 Design and Implementation Constraints

- **Technology Stack:** MERN stack (MongoDB, Express.js, React.js, Node.js).
 - **Scalability:** Multi-tenant architecture with `collegeId` in all core collections.
 - **Security:** HTTPS, strong password hashing, JWT for authentication, and **time-sensitive OTPs** for delivery verification.
 - **Payment Gateway:** Integration with a PCI-compliant payment gateway.
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3. System Features (Functional Requirements)

3.1 User Authentication & Management

- **FR-1.1:** Unified registration page requiring PRN (Student) or Teacher ID (Faculty).
- **FR-1.2:** Unified login page authenticating users via PRN/Teacher ID/Admin Username.
- **FR-1.3:** Session management via JWT.
- **FR-1.4:** “Forgot Password” functionality via email.

3.2 Student & Faculty Features

- **FR-2.1 (Dashboard):** Personalized dashboard with quick access to all module features.
- **FR-2.2 (Canteen Ordering & Delivery):**
 - Users can browse the menu and add items to a cart.
 - Users must specify a **Delivery Location** (e.g., Hostel Block A, Classroom 301).
 - Users can view and apply available **Promotional Offers** to their cart.
 - Checkout options: “Pay Online” or “Pay with Cash.”
 - **Delivery is free of charge.**
- **FR-2.3 (Order Tracking & OTP):**
 - Users receive real-time status updates (Confirmed, Preparing, Out for Delivery).
 - When the status changes to “**Out for Delivery,**” a unique, time-sensitive **4-digit OTP** is generated and displayed to the user.
 - The user must provide this OTP to the delivery person upon receipt.
- **FR-2.4 (Mess Voting):** Users can cast one vote per active mess menu poll.
- **FR-2.5 (Mess QR Code):** Unique QR code for meal redemption.
- **FR-2.6 (Transaction History):** Complete history of orders, including applied offers and final payment status.

3.3 Canteen Administrator Features

- **FR-3.1 (Order Dashboard):** Real-time dashboard displaying incoming orders, including delivery location and payment status.
- **FR-3.2 (Order Management & Verification):**
 - Admin can update order status (Confirmed, Preparing, Out for Delivery).
 - When marking an order as “**Delivered,**” the admin must input the **OTP** provided by the customer for verification.
- **FR-3.3 (Menu Management):** Admin can add, edit, delete, and categorize food items.
- **FR-3.4 (Promotional Offer Management):**
 - Admin can create, activate, and deactivate promotional offers (e.g., percentage discount, fixed amount off, minimum order value).
 - Offers must have a start date, end date, and usage limits.
- **FR-3.5 (Financial Reporting):** Reports on revenue, payment breakdown, and offers utilized.

3.4 Mess Administrator Features

- **FR-4.1 (Voting Management):** Create, manage, and close voting polls.
- **FR-4.2 (Voting Analytics):** View real-time voting results.
- **FR-4.3 (Menu Publishing):** Finalize and publish the official weekly menu.
- **FR-4.4 (Meal Redemption):** Scan user QR codes to validate and record meal redemptions.

3.5 Food Wastage Tracker

- **FR-5.1:** Staff can input the weight of food wasted after each meal service.
 - **FR-5.2:** Public-facing dashboard displays daily and historical wastage trends.
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4. External Interface Requirements

4.1 User Interfaces (UI)

- **Technology:** Responsive SPA built with **React.js**.
- **Real-time:** **Socket.IO** for order status updates and OTP display.

4.2 Software Interfaces (API)

- **Technology:** RESTful API built with **Node.js** and **Express.js**.
- **Endpoints:** Dedicated endpoints for authentication, ordering, administration, and payment webhooks.

4.3 Hardware Interfaces

- User devices (smartphones/laptops) for ordering.
- Mess/Canteen staff devices (tablets/smartphones) for QR scanning and **OTP input**.

4.4 Communications Interfaces

- All client-server communication over **HTTPS**.
 - Real-time communication via **Socket.IO** (WebSockets).
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5. Non-Functional Requirements

5.1 Performance

- API response times for critical requests under 500ms.
- Initial load time under 3 seconds.

5.2 Scalability

- Multi-tenant architecture (`collegeId` in all core collections).
- Stateless backend for horizontal scaling.

5.3 Security

- Protection against XSS, CSRF, and injection attacks.
- Strict Role-Based Access Control (RBAC).
- **OTP Security:** OTPs must be time-sensitive (e.g., valid for 5 minutes) and stored securely (hashed or encrypted) in the database until used or expired.

5.4 Reliability

- 99.9% uptime target.
- Robust error handling and logging.

5.5 Usability

- Intuitive UI requiring minimal training.
 - Clear visual distinction for applied offers and final order amount.
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6. System Architecture & Data Flow

6.1 Technology Stack

- **Frontend:** React.js
- **Backend:** Node.js, Express.js
- **Database:** MongoDB
- **Real-time Engine:** Socket.IO

6.2 API Endpoint Specification

Category	Method	Endpoint	Description
Auth	POST	/api/auth/register	Register Student/Faculty (requires PRN/Teacher ID)
Auth	POST	/api/auth/login	Authenticate user (via PRN/Teacher ID/Admin Username)
Menu	GET	/api/canteen/items	Get all available canteen items
Offers	GET	/api/offers	Get all active promotional offers
Offers	POST	/api/admin/offers	Admin creates a new offer
Order	POST	/api/orders/canteen	Place a new canteen order (includes delivery location, applied offer)
Order	GET	/api/orders/me	Get user's order history
Admin	GET	/api/admin/orders	Admin views all active orders
Admin	PUT	/api/admin/orders/:id/status	Admin updates order status (e.g., "Out for Delivery")
Admin	POST	/api/admin/orders/:id/verify	Admin verifies delivery using OTP
Payment	POST	/api/payments/initiate	Initiate online payment (sends final amount after offer)
Payment	POST	/api/payments/webhook	Webhook from Payment Gateway to confirm transaction
Mess	POST	/api/mess/vote	User casts a vote for the menu
Mess	GET	/api/mess/qrcode	Get user's meal redemption QR code data

6.3 Database Schema

Collection	Key Fields Added/Modified	Description
users	prn , teacherId , role , collegeId	User profiles.
items	collegeId , price , isAvailable	Canteen food items.
offers	collegeId , code , type , value , minOrder , startDate , endDate	New collection for promotional offers.
orders	collegeId , deliveryLocation , appliedOfferCode , discountAmount , finalAmount , otp , otpExpiresAt	Order details, now including delivery and offer information, and the OTP .
polls	collegeId , votes	Mess menu voting polls.

6.4 Payment Gateway Integration Flow

The flow remains the same, but the amount passed to the gateway in the `initiate` step must be the `finalAmount` after the discount from any applied offer has been deducted.

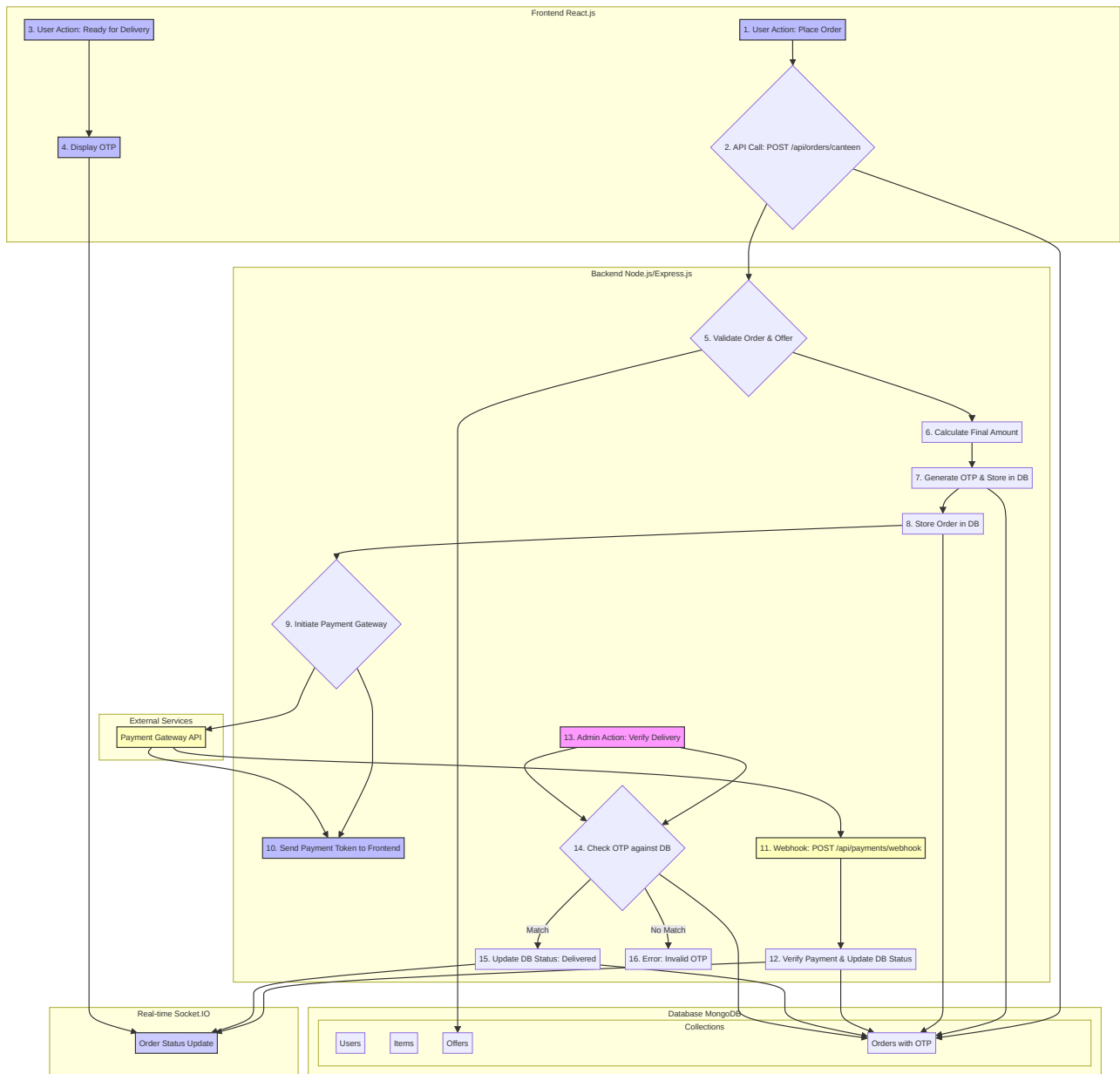
7. Future Enhancements (Real-World Features)

- **Inventory Management:** Track ingredient stock levels and automate reorder alerts.
 - **Dynamic Pricing:** Implement time-based discounts (e.g., Happy Hour).
 - **Feedback & Rating System:** Allow users to rate food items and delivery service.
 - **Advanced Analytics:** Comprehensive dashboard for sales trends, peak hours, and offer performance.
 - **Subscription Plans:** Allow students to subscribe to weekly or monthly mess plans.
 - **Third-Party Delivery Integration:** Option to integrate with campus internal delivery staff management or external services.
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API Connection and Data Flow Visualization (Updated)

The core architecture remains a MERN stack, but the flow now includes the **Offers** and **OTP** logic.

Visual Representation of Data Flow



Key Flow Changes:

1. **Order Placement (Steps 5-8):** The backend now includes steps to **Validate the Offer** (against the `offers` collection), **Calculate the Final Amount**, and **Generate and Store the OTP** in the `orders` collection before payment initiation.

2. **Delivery Verification (Steps 13-15):** The Canteen Admin's action to mark an order as "Delivered" is now a dedicated API call (`POST /api/admin/orders/:id/verify`) that requires the **OTP** as a parameter. The backend verifies this OTP against the one stored in the database.

This completes the updated SRS and the visualization. I will now deliver the file.